ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

TECHNICAL MATHEMATICS P1

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Technical Mathematics Paper 1 practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept correct alternative methods; give follow-through marks (CA) where an earlier error is used correctly.

QUESTION 1

1.1 $\sqrt{9} = 3$: rational ✓; π: irrational ✓; $-\frac{3}{4}$: rational ✓; $\sqrt{-4}$: non-real ✓ (4)

1.2 $45 = 32 + 8 + 4 + 1$ ✓ $= 101101_2$ ✓ (2)

1.3 $16 + 8 + 0 + 2 + 0$ ✓ $= 26$ ✓ (2)

1.4 $1011_2 + 110_2 = 10001_2$ ✓✓ $(11 + 6 = 17)$ (2)

1.5 $3{,}142$ ✓ (1)

1.6 $4{,}5$ ✓ $\times 10^{-5}$ ✓ (2)

1.7 $\sqrt{49} = 7$ and $\sqrt{64} = 8$, so between 7 ✓ and 8 ✓ (2)

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QUESTION 2

2.1.1 $8a^9b^6$ ✓✓ (2)

2.1.2 $= \frac{x^3}{x^2}$ ✓✓ $= x$ ✓ (3)

2.1.3 $4x^4$ ✓✓ (2)

2.1.4 $2\sqrt{(4 \times 3)} + \sqrt{(9 \times 3)} = 4\sqrt{3}$ ✓ $+ 3\sqrt{3}$ ✓ $= 7\sqrt{3}$ ✓ (3)

2.2.1 $(x - 3)(x - 4)$ ✓✓ (2)

2.2.2 $(2x - 5)(2x + 5)$ ✓✓ (2)

2.2.3 $3x(x + 2)$ ✓✓ (2)

2.3 $= \frac{(x - 2)(x + 2)}{x + 2}$ ✓✓ $= x - 2$ ✓ (3)

2.4 $2x^2 - 3x + 8x - 12$ ✓✓ $= 2x^2 + 5x - 12$ ✓ (3)

2.5 $6x - 3 - 2x - 8$ ✓✓ $= 4x - 11$ ✓ (3)

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QUESTION 3

3.1.1 $3x = 12$ ✓ $\rightarrow x = 4$ ✓ (2)

3.1.2 $(x - 3)(x + 2) = 0$ ✓ $\rightarrow x = 3$ ✓ or $x = -2$ ✓ (3)

3.1.3 $2^x = 2^5$ ✓ $\rightarrow x = 5$ ✓ (2)

3.2 $-2x > 4$ ✓ $\rightarrow x < -2$ ✓ (sign reverses when dividing by a negative); number line with an open circle at -2 and arrow to the left ✓ (3)

3.3 $3x + (2x + 1) = 11$ ✓ $\rightarrow 5x = 10$ ✓ $\rightarrow x = 2$ ✓; $y = 2(2) + 1 = 5$ ✓ (4)

3.4 $R = \frac{V}{I}$ ✓ $= \frac{12}{0,5}$ ✓ $= 24\ \Omega$ ✓ (3)

3.5 $30 = 2(9 + b)$ ✓ $\rightarrow 15 = 9 + b$ ✓ $\rightarrow b = 6$ cm ✓ (3)

3.6 Let width $= x$: $x(x + 3) = 40$ ✓ $\rightarrow x^2 + 3x - 40 = 0$ ✓ $\rightarrow (x + 8)(x - 5) = 0$ ✓ (5)
$x = 5$ (width cannot be negative) ✓; width 5 cm and length 8 cm ✓

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QUESTION 4

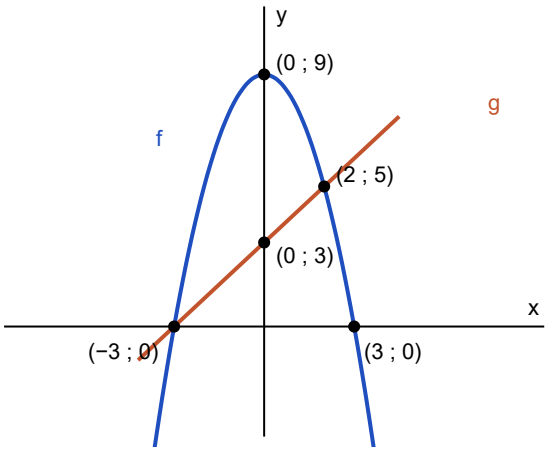
4.1 $(0\ ;\ 9)$ ✓ (1)

4.2 $-x^2 + 9 = 0 \rightarrow x = \pm 3$ ✓; $(-3\ ;\ 0)$ and $(3\ ;\ 0)$ ✓ (2)

4.3 $x + 3 = 0 \rightarrow (-3\ ;\ 0)$ ✓ (1)

4.4 $x = 0$ ✓; gradient $= 1$ ✓ (2)

4.5 (5)



Marks: shape of *f* (opens downwards) ✓; turning point (0 ; 9) ✓; *x*-intercepts of *f* ✓; straight line *g* with positive gradient ✓; intercepts of *g*: (0 ; 3) and (−3 ; 0) ✓

4.6 $-x^2 + 9 = x + 3 \rightarrow x^2 + x - 6 = 0$ ✓ $\rightarrow (x + 3)(x - 2) = 0$ ✓ (4)
$(-3\ ;\ 0)$ ✓ and $(2\ ;\ 5)$ ✓

4.7 $y \leq 9$ ✓ (1)

4.8 $-3 \leq x \leq 2$ ✓✓ (2)

4.9 $x = 0$ ✓ and $y = 0$ ✓; quadrants 1 and 3 ✓ (3)

4.10 $(0\ ;\ 1)$ ✓; $y = 0$ ✓ (2)

4.11 $y = x - 2$ ✓✓ (2)

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QUESTION 5

5.1 (a) $A = P(1 + i)^n = 15\,000(1{,}09)^4$ ✓✓ $= R21\,173{,}72$ ✓✓ (7)

(b) $A = P(1 + in) = 15\,000(1 + 0{,}09 \times 4)$ ✓✓ $= R20\,400$ ✓

5.2 $21\,173{,}72 - 20\,400 = R773{,}72$ ✓ (1)

5.3 $A = 40\,000(1 - 0{,}1 \times 3)$ ✓ $= R28\,000$ ✓ (2)

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TOTAL: 100

END OF MARKING GUIDELINE